

RANK of a Matrix

Rank:

The number of non-zero rows in REF is called rank of the matrix.

Or

The number of pivots in REF is called rank of the matrix.

Example:

$$\begin{pmatrix} 1 & 2 & -1 & 0 \\ 0 & 3 & 4 & 1/2 \\ 0 & 0 & -44 & -4 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad \begin{matrix} \text{Rank} = 3 \\ \text{REF} \end{matrix}$$

$$\begin{pmatrix} 1 & 2 & -1 & 0 \\ 0 & 3 & 4 & 1/2 \\ 0 & 0 & -44 & -4 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad \begin{matrix} \text{Rank} = 3 \\ \text{REF} \end{matrix}$$

PROPERTIES OF RANK OF A MATRIX

- Rank of a null matrix/zero matrix is always zero.
- Rank of any non-zero matrix is always greater than or equal to 1.
- $\text{rank}(A_{m \times n}) \leq \min(m, n)$

Example: $\text{rank}(A_{5 \times 3}) \leq \min(5, 3)$

- Let $A = (a_{i,j})_{n \times n}$ be any non-singular matrix then $\text{rank}(A) = n$.
- $\text{rank}(A) = \text{rank}(A^T)$. i.e., Rank of transpose of matrix A is always equal to rank of A.
- The column rank of a matrix is always equal to its row rank
- Rank of product of two matrices cannot exceed the rank of both of the matrices.

Example

$$\begin{aligned}
 & \boxed{1.} \quad \begin{pmatrix} 2 & -3 & 1 & 2 \\ 4 & -6 & 2 & 4 \\ -4 & 6 & -1 & 0 \end{pmatrix} \\
 & \sim \begin{pmatrix} 2 & -3 & 1 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 4 \end{pmatrix} \quad \begin{array}{l} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 + R_2 \end{array} \\
 & \sim \begin{pmatrix} 2 & -3 & 1 & 2 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad \begin{array}{l} R_2 \leftrightarrow R_3 \\ \text{REF} \end{array} \\
 & \sim \begin{pmatrix} 2 & -3 & 0 & -2 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad R_1 \rightarrow R_1 - R_2 \\
 & \sim \begin{pmatrix} 1 & -3/2 & 0 & -1 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad \begin{array}{l} R_1 \rightarrow \frac{R_1}{2} \\ \text{RREF} \end{array}
 \end{aligned}$$

$$\begin{aligned}
 & \sim \begin{pmatrix} 1 & -\frac{3}{2} & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad C_4 \rightarrow C_4 - 4C_3 \\
 & \sim \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad \begin{array}{l} C_4 \rightarrow C_4 + C_1 \\ C_2 \rightarrow C_2 + \frac{3}{2}C_1 \end{array} \\
 & \sim \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad C_3 \leftrightarrow C_2 \\
 & \sim \begin{pmatrix} I_2 & \vdots & 0 \\ \vdots & \vdots & \vdots \\ 0 & \vdots & 0 \end{pmatrix} \quad \text{NF}
 \end{aligned}$$

Rank=2

Example

$$\begin{aligned}
 & \boxed{2.} \quad \begin{pmatrix} -2 & 1 & 1 \\ 0 & 8 & -4 \\ 3 & 2 & -5 \end{pmatrix} \\
 & \sim \begin{pmatrix} -2 & 1 & 1 \\ 0 & 8 & -4 \\ 0 & 7 & -7 \end{pmatrix} \quad R_3 \rightarrow 2R_3 + 3R_1 \\
 & \sim \begin{pmatrix} -2 & 1 & 1 \\ 0 & 2 & -1 \\ 0 & 1 & -1 \end{pmatrix} \quad \begin{array}{l} R_2 \rightarrow \frac{R_2}{4} \\ R_3 \rightarrow \frac{R_3}{7} \end{array} \\
 & \sim \begin{pmatrix} -2 & 1 & 1 \\ 0 & 2 & -1 \\ 0 & 0 & 1 \end{pmatrix} \quad R_3 \rightarrow R_2 - 2R_3 \\
 & \quad \text{REF}
 \end{aligned}$$

$$\begin{aligned}
 & \sim \begin{pmatrix} -2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \begin{array}{l} R_1 \rightarrow R_1 - R_3 \\ R_2 \rightarrow R_2 + R_3 \end{array} \\
 & \sim \begin{pmatrix} -4 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad R_1 \rightarrow 2R_1 - R_2 \\
 & \sim \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \begin{array}{l} R_1 \rightarrow \frac{R_1}{-4} \\ R_2 \rightarrow \frac{R_2}{2} \end{array} \\
 & \quad \text{RREF} \\
 & \sim (I_3) \quad \text{NF}
 \end{aligned}$$

Rank=3

Exercise

1. Short questions:

- What are the possible normal forms of matrices?

- ii. How many REF, RREF and NF do exist for a matrix?
- iii. When will you have RREF=NF?
- iv. Why can't a matrix have 2 different normal forms?
- v. What are possible normal forms of 2×3 and 2×2 matrices.
- vi. Write a 5×4 matrix for which $EF=RREF=NF$.
- vii. Which matrices do not possess any normal form?
- viii. What is the normal form of a zero matrix?(ans. undefined)
- ix. What is the rank of a 23×27 zero matrix?
- x. What is the rank of $-233I_{137}$?
- xi. What is the rank of a permutation matrix of order 27×27 ?
- xii. What maybe the maximum & minimum ranks of a 27×37 non-zero matrix?
- xiii. M is 25×15 & $\text{rank}(M)=15$. What will be its RREF & NF?
- xiv. M is 4×8 & $\text{rank}(M)=3$. What is it: True or False?
- xv. A rank deficient matrix has dimension 7×9 . Comment on its rank.
- xvi. L is 1×29 , M is 29×1 ; one of them is a null matrix. Find rank (ML) .
- xvii. L is 1×29 , M is 29×1 ; both are non-zero matrices. Find rank (LM) .
- xviii. S is a sub-matrix of M . Which is true among the followings?

$$\text{Rank}(S) < \text{rank}(M);$$

$$\text{rank}(S) > \text{rank}(M);$$

$$\text{rank}(S) \geq \text{rank}(M);$$

$$\text{rank}(S) \leq \text{rank}(M)$$

2. Find Rank, RREF and NF for the followings:

$$1. \begin{bmatrix} 3 & 2 & 1 & 3 & 0 \\ -1 & 2 & 4 & 6 & 7 \\ 0 & 2 & 2 & 4 & 4 \end{bmatrix}$$

$$2. \begin{bmatrix} 4 & 0 & 0 \\ 1 & 3 & -1 \\ 2 & 2 & 5 \end{bmatrix}$$

$$3. \begin{bmatrix} 4 & -2 & 8 & 0 & 4 \\ 1 & -1 & 9 & 0 & -5 \\ 0 & -7 & 5 & 0 & 2 \end{bmatrix}$$

$$4. \begin{bmatrix} 3 & 0 & 1 & -3 & 0 \\ -1 & 5 & 4 & 4 & 9 \\ 0 & 0 & 2 & 4 & -1 \end{bmatrix}$$

$$5. \begin{bmatrix} 1 & 3 & -2 & 0 & 2 & 0 \\ 2 & 6 & -5 & -2 & 4 & -3 \\ 8 & 0 & 5 & 10 & 0 & 15 \\ 2 & 6 & 0 & 8 & 4 & 18 \end{bmatrix}$$

$$6. \begin{bmatrix} -1 & 3 & 1 \\ 7 & 5 & 6 \\ 2 & -2 & 0 \\ 4 & 2 & 3 \\ 3 & -9 & -3 \end{bmatrix}$$

$$7. \begin{bmatrix} 3 & -2 & 0 & -1 \\ 0 & 2 & 2 & 1 \\ 1 & -2 & -3 & 2 \\ 0 & 1 & 2 & 1 \end{bmatrix}$$

$$8. \begin{bmatrix} 1 & 2 & -3 & -2 & -3 \\ 1 & 3 & -2 & 0 & -4 \\ 3 & 8 & -7 & -2 & -11 \\ 2 & 1 & -9 & -10 & -3 \end{bmatrix}$$

$$9. \begin{bmatrix} 0 & -1 & -1 & 0 & 2 & 4 \\ 2 & 2 & 4 & -2 & 4 & -6 \\ 1 & 0 & 1 & 10 & 0 & 10 \\ 3 & 4 & 7 & 8 & 4 & 16 \end{bmatrix}$$

$$10. \begin{bmatrix} 1 & -1 & 2 & 1 \\ 3 & 0 & 2 & 2 \\ 2 & 1 & -1 & 1 \\ 1 & 0 & 1 & 1 \end{bmatrix}$$